

PORTO AP, Portugal

WMO: 085450

Lat: 41.2322N

Lon: 8.6792W

Elev: 70

StdP: 100.49

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.3	3.6	-5.4	2.4	8.4	-3.2	2.9	8.3	11.1	13.0	9.8	12.6	2.2	100	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.6	30.5	18.7	28.3	18.0	26.3	17.5	20.4	25.9	19.6	24.4	19.0	23.2	3.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.9	21.2	18.2	13.2	20.5	17.8	12.9	20.2	58.8	25.8	56.1	24.5	54.2	23.1	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.3	8.2	7.2	DB	-0.2	34.8	1.2	2.2	-1.1	36.4	-1.8	37.7	-2.4	38.9	-3.3	40.4	
			WB	-1.7	22.5	1.4	1.5	-2.7	23.6	-3.5	24.5	-4.3	25.4	-5.3	26.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	10.3	10.7	12.7	13.8	15.8	18.3	19.4	19.9	18.9	16.9	12.9	11.1
	DBStd	4.35	2.48	2.51	2.72	2.87	2.92	2.90	2.47	2.60	2.60	2.74	2.73	2.54
	HDD10.0	78	27	19	6	2	0	0	0	0	0	0	6	18
	HDD18.3	1393	249	213	177	141	92	33	11	7	21	62	163	226
	CDD10.0	1932	36	39	88	116	180	249	292	307	267	213	93	52
	CDD18.3	205	0	0	1	4	13	32	44	55	38	17	1	0
	CDH23.3	1296	0	0	11	26	94	241	299	366	192	67	1	0
	CDH26.7	387	0	0	0	3	22	75	104	127	47	9	0	0
Wind	WSAvg	3.1	3.2	3.2	3.4	3.3	3.3	3.1	2.9	2.7	2.6	3.0	3.1	3.3
Precipitation	PrecAvg	1252	166	148	100	114	94	48	16	24	67	131	164	183
	PrecMax	2005	425	386	365	400	233	253	72	89	193	349	484	624
	PrecMin	860	4	17	0	11	2	1	0	0	1	12	15	27
	PrecStd	246	98	101	83	82	60	47	18	22	49	87	103	134
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	20.4	24.8	26.2	29.4	32.0	32.9	33.6	30.7	28.0	22.1	18.6
		MCWB	12.3	12.8	13.4	16.2	17.5	18.6	19.6	19.4	17.9	17.4	15.0	12.7
	2%	DB	15.8	17.6	21.8	23.1	26.0	28.9	29.6	30.1	27.9	25.1	19.8	16.9
		MCWB	12.3	11.7	12.5	14.7	16.6	17.8	19.1	18.8	17.4	16.6	14.6	13.1
	5%	DB	14.9	15.8	19.1	20.8	23.1	26.1	26.7	27.2	25.4	23.0	18.2	15.9
		MCWB	12.4	11.4	12.3	14.2	15.9	17.2	18.6	18.0	17.1	16.2	14.8	13.4
	10%	DB	14.1	14.6	17.0	18.6	20.7	23.5	23.9	24.8	23.4	21.0	17.1	15.0
		MCWB	12.1	11.5	11.6	13.4	15.2	17.0	18.2	17.8	17.0	16.2	14.2	12.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	14.8	15.4	17.7	19.4	20.8	21.5	21.9	20.2	19.9	18.1	16.0
		MCDB	15.6	16.5	19.6	23.1	25.3	26.9	28.3	27.4	25.0	22.6	18.8	16.3
	2%	WB	14.0	13.8	14.4	16.0	17.9	19.6	20.3	20.2	19.2	18.9	16.7	15.0
		MCDB	14.7	15.3	17.7	20.4	22.9	25.4	26.3	25.0	23.3	21.4	18.0	15.7
	5%	WB	13.3	13.1	13.7	15.0	16.8	18.6	19.4	19.5	18.6	18.1	15.8	14.1
		MCDB	14.2	14.5	16.5	18.8	21.3	23.4	24.3	23.8	22.3	20.6	17.3	15.2
	10%	WB	12.7	12.3	13.0	14.2	16.0	17.8	18.7	18.8	18.0	17.4	14.9	13.2
		MCDB	13.8	14.0	15.6	17.6	19.4	21.9	22.7	22.8	21.5	19.9	16.5	14.6
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.6	7.9	7.8	7.5	7.9	8.0	8.6	8.4	7.7	7.2	6.7
		MCDBR	7.1	9.2	11.8	11.4	11.5	12.6	12.8	12.9	11.7	10.7	8.1	6.9
		MCWBR	5.0	5.5	6.4	6.0	5.7	5.7	5.6	5.4	5.0	5.0	5.0	4.8
	5% WB	MCDBR	4.5	6.4	8.7	9.4	9.1	9.7	9.8	9.4	8.5	7.2	5.4	5.0
		MCWBR	4.2	5.0	5.5	5.5	5.0	5.0	5.0	4.8	4.5	4.3	4.4	4.6
Clear-Sky Solar Irradiance	taub	0.332	0.342	0.375	0.374	0.382	0.385	0.375	0.387	0.376	0.368	0.340	0.328	
	taud	2.503	2.465	2.380	2.391	2.395	2.402	2.420	2.384	2.437	2.475	2.509	2.510	
	Ebn at Noon	820	868	873	897	895	889	894	873	861	826	804	793	
	Edh at Noon	75	90	109	115	117	117	114	114	102	88	74	69	
All-Sky Solar Radiation	RadAvg	1.84	2.87	3.96	5.24	6.39	6.96	7.09	6.41	4.99	3.22	2.10	1.67	
	RadStd	0.31	0.40	0.58	0.59	0.56	0.46	0.36	0.34	0.36	0.38	0.36	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.33	N/A	N/A	N/A	-0.31	N/A	N/A	N/A	+108	+40
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+39

Nomenclature: See separate page